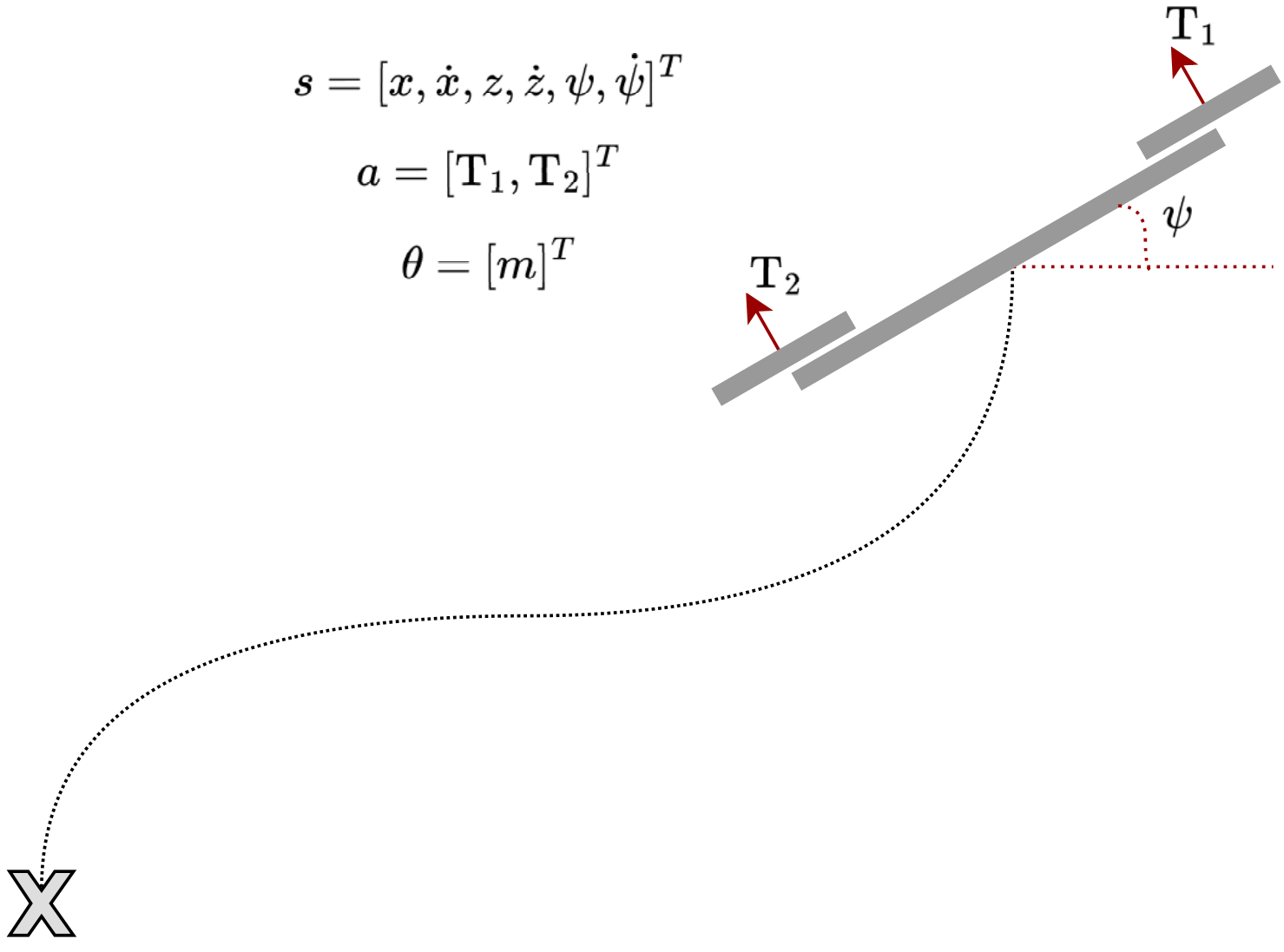
$$s = [x, \dot{x}, z, \dot{z}, \psi, \dot{\psi}]^T$$

$$a = [\mathrm{T}_1, \mathrm{T}_2]^T$$

$$\theta = [m]^T$$